

HAORAN JIANG

Tokyo, Japan

kyokouzen9711@gmail.com ◇ [ORCID](#) ◇ [Google Scholar](#)

EDUCATION

Doctor of Philosophy in Engineering, University of Tsukuba, Japan

Apr. 2021 – Mar. 2024

Supervisor: Prof. Takashi Matsushima

- Developed a level set discrete element method (LS-DEM) framework to model particle degradation (abrasion and fragmentation) and investigated its influence on the mechanical behavior of granular materials.
- Developed image-based particle reconstruction techniques using optical imaging and spherical harmonics, and investigated the influence of particle morphology and surface roughness on the macro- and micro-mechanical behavior of granular materials.
- Quantified mesoscale structures (cells and clusters) in granular systems by integrating DEM simulations with statistical mechanics, providing insights into constitutive modeling.
- Proposed an improved rheological model for bidisperse granular flows and clarified the mechanisms of particle-size segregation.

Master of Engineering, University of Tsukuba, Japan

Apr. 2019 – Mar. 2021

Supervisor: Prof. Takashi Matsushima

Bachelor of Engineering, Wuhan University of Technology, China

Sep. 2014 – Jun. 2018

Supervisor: Prof. Bin Huang

PROFESSIONAL EXPERIENCE

Kajima Corporation, Japan

Apr. 2024 – Present

Technical Chief, Advanced Analysis Department, Civil Engineering Design Division

- Developed a coupled DEM–FEM framework to simulate crack propagation in full-scale reinforced concrete (RC) structures and investigate the relationship between local reinforcement strain and concrete damage.
- Conducted coupled DEM–FVM simulations of underwater filling behavior of bentonite pellets for geological disposal systems, aiming to optimize sealing performance.
- Developed a coupled DEM–SPH framework to investigate the flowability and clogging mechanisms of fresh concrete considering realistic aggregate size distributions and particle morphologies.
- Established a DEM model for foamed soil and applied it to evaluate excavation face stability in shield tunneling.

Technical Skills: LS-DEM, MELODY, SudoSim, PFC3D, LIGGGHTS, OpenFOAM, FLAC3D, DualSPHysics, GeoTaichi, MATLAB, C++, Fortran.

PUBLICATIONS

Journal Papers

1. **Haoran Jiang***, Ruidong Li, Xiaoyu Jiang, & Quanshui Huang (2026). Identification and quantification of orientationally ordered clusters in packings of frictionless elliptical particles (**PARTICUOLOGY Featured Article**). *Particuology*, 118, 217–229.
2. Xuan Huang, **Haoran Jiang***, Quanshui Huang, Bo Xu, Bin Ou, & Xing Zhou (2026). New insight into mechanism of particle size effect: synergy of microstructural heterogeneity and contact conditions. *Rock Mechanics and Rock Engineering*, in press.
3. **Haoran Jiang**, Aoxi Zhang, Chaofa Zhao, & Zhongxuan Yang (2026). A new approach for modeling granular materials with complex particle shapes using clumps of convex bodies. *Powder Technology*, 483, 122881.

4. **Haoran Jiang**, Jiaying Liu, Zi Ye, & Yang Li (2026). Cell-based origins of structural and mechanical responses in sheared granular systems. *Computers and Geotechnics*, 197, 108216.
5. **Haoran Jiang**, Ning Guo, & Fan Chen (2026). Shearing rigid–soft granular mixtures: Intrinsic micro–meso signatures underlying critical-state strength. *International Journal of Solids and Structures*, 334, 113997.
6. **Haoran Jiang***, Hiroyuki Shimizu, Shintaro Ohno, & Takayuki Sogawa (2026). DEM-based modeling framework for steel-reinforced concrete structures: A case study of four-point bending test. *Powder Technology*, 474, 122253.
7. Ruidong Li, **Haoran Jiang***, Brian Sheil, Chengkai Xu, & Shao-heng He* (2026). Effects of particle morphology and intermediate principal stress ratio on the multiscale granular behavior: Insights from μ CT-informed LS-DEM modeling. *Canadian Geotechnical Journal*, 63, 1–27.
8. Quanshui Huang, Xuan Huang, Huanfeng Qiu, **Haoran Jiang**, Bo Xu, & Xing Zhou (2026). An insight into the competition between the shape effects and grain crushing behavior: From single grain to aggregates. *Acta Geotechnica*, 21, 4229–4248.
9. **Haoran Jiang**, Yang Li, Quanshui Huang, Dawei Xue, & Giuseppe Buscarnera (2026). Numerical identification of size–shape–strength attractors in sheared breakable granular materials. *Computers and Geotechnics*, 190, 107709.
10. Qi Zhang, **Haoran Jiang**, Gujie Sang, & Yike Dang (2025). The shakedown behaviour of binary soil: A DEM study. *Particuology*, 108, 196–206.
11. Dominik Krenzel, **Haoran Jiang**, Takashi Matsushima, & Raphael Blumenfeld (2025). Effects of particle angularity on granular self-organization. *Physical Review E*, 112, 055407.
12. **Haoran Jiang**, Opu Chandra Debanath, Yang Li, & Ruidong Li (2025). A level-set method-based framework for modeling abrasion of railway ballast. *Transportation Geotechnics*, 54, 101628.
13. Ruidong Li, Shao-heng He, **Haoran Jiang**, Chengkai Xu, & Ningyu Yang (2025). Morphology-controlled single rock particle breakage: A finite-discrete element method study with fractal dimension analysis. *Fractal and Fractional*, 9, 562.
14. Yang Li, & **Haoran Jiang*** (2025). Investigating rotational characteristics and contact mechanisms of star-like shapes using multiellipse-based DEM. *Particuology*, 99, 210–225.
15. **Haoran Jiang**, Jiayan Nie, Opu Chandra Debanath, & Yang Li (2025). Dynamic column collapse of dry granular materials with multi-scale shape characteristics. *Computers and Geotechnics*, 177, 106873.
16. Dominik Krenzel, **Haoran Jiang**, Jian Chen, & Takashi Matsushima (2025). The combined effect of particle angularity and inter-particle friction on micro- and macroscopic properties of granular assemblies. *Computers and Geotechnics*, 177, 106850.
17. Yang Li, Yang Dong, **Haoran Jiang***, & Zhenming Shi (2024). Exploring the micromechanical origin of shear response in granular materials induced by size non-uniformity. *Granular Matter*, 26, 100.
18. **Haoran Jiang**, Opu Chandra Debanath, Fan Chen, Reid Kawamoto, & Yang Li (2024). Analyzing the mechanical behavior of granular materials: A multi-morphological approach using spherical harmonics and LS-DEM. *Powder Technology*, 445, 120078.
19. **Haoran Jiang**, Reid Kawamoto, & Yang Li (2024). Shape-induced clusters of ellipsoids during triaxial compression: A multiscale analysis using LS-DEM. *Computers and Geotechnics*, 169, 106235.
20. **Haoran Jiang**, Reid Kawamoto, & Takashi Matsushima (2024). Coupled effects of particle shape and inter-particle friction on quasi-static shear behavior of dry granular materials studied by 2D LS-DEM. *Journal of Structural Engineering*, A, 70, 141–154.
21. **Haoran Jiang**, Xiaoyu Jiang, & Takashi Matsushima (2023). Rheology of segregated bi-disperse granular flow in an inclined plane. *Journal of JSCE*, 11, 22–15035.

Conference Papers

1. Zhipeng Yu, Yang Li, & **Haoran Jiang*** (2025). Biaxial shearing of star-shaped particles with multiellipse DEM model. *EPJ Web of Conferences*, 340, 09001.
2. **Haoran Jiang**, Reid Kawamoto, & Takashi Matsushima (2024). Exploring the combined effect of particle shape and friction on cell structures in granular packings via LS-DEM modeling. *IOP Conference Series: Earth and Environmental Science*, 1330, 012047.
3. **Haoran Jiang**, Opu Chandra Debanath, Reid Kawamoto, Takashi Matsushima, Shuichi Adachi, & Masahiro Miwa (2023). Calibration of railway ballast modeling using level set discrete element method. In *International Conference on Computational & Experimental Engineering and Sciences*.
4. **Haoran Jiang**, Reid Kawamoto, & Takashi Matsushima (2022). Shape effects on shear behavior of superdisk systems studied by level set discrete element method. In *International Conference on Advances in Civil Engineering*.

* Corresponding author.

CONFERENCE PRESENTATIONS

1. **Haoran Jiang**, Hiroyuki Shimizu, & Shintaro Ohno. A practical FEM–DEM coupling strategy for capturing the cracking propagation of reinforced concrete structures. *The 10th International Conference on Discrete Element Methods (DEM10)*, Japan, Jul. 2025.
2. **Haoran Jiang**, Reid Kawamoto, & Takashi Matsushima. Shape-dependent rheology of dry granular flows studied by LS-DEM. *The 13th International Conference on Geotechnique, Construction Materials and Environment*, Japan, Nov. 2023.
3. **Haoran Jiang**, Reid Kawamoto, & Takashi Matsushima. Cell-based microscopic insights into granular packings: Combined effects of particle shape and inter-particle friction. *33rd ALERT Geomaterials Workshop and School*, France, Sep. 2023.
4. **Haoran Jiang**, Opu Chandra Debanath, Reid Kawamoto, Takashi Matsushima, Shuichi Adachi, & Masahiro Miwa. Calibration of railway ballast modeling using level set discrete element method. *29th International Conference on Computational & Experimental Engineering and Sciences (ICCES2023)*, China, May 2023.
5. **Haoran Jiang**, & Takashi Matsushima. DEM–SPH coupling for simulating complex erosion and sedimentation processes. *Tsukuba Global Science Week (TGSW2022)*, Japan, Sep. 2022.
6. **Haoran Jiang**, Xiaoyu Jiang, & Takashi Matsushima. Bi-disperse granular flow down an inclined plane studied by 2D discrete element simulations. *15th World Congress on Computational Mechanics & 8th Asian Pacific Congress on Computational Mechanics (WCCM–APCOM 2022)*, Japan, Aug. 2022.

HONORS AND AWARDS

- Outstanding Presentation Award, 60th Japanese Geotechnical Engineering Annual Conference, Japanese Geotechnical Society, Jul. 2025.
- Best Student Paper Award, *Calibration of Railway Ballast Modeling Using Level Set Discrete Element Method*, 29th International Conference on Computational & Experimental Engineering and Sciences (ICCES2023), May 2023.
- Outstanding Presentation Award, 25th Symposium on Applied Mechanics, Japan Society of Civil Engineers, May 2022.
- Outstanding Presentation Award, GeoKanto 2021, Kanto Branch of the Japanese Geotechnical Society, Oct. 2021.

PROFESSIONAL SERVICE

Committee Membership

- Editorial Member, *Journal of JGS*, Japanese Geotechnical Society.
- Committee Member, Subcommittee on Coupled Phenomena in Rock Engineering, Japan Society of Civil Engineers.
- Committee Member, Subcommittee on Seismic Assessment and Strengthening of Historic Castle Stone Walls and Masonry Retaining Walls, Japan Society of Civil Engineers.
- Committee Member, Research Committee on Multi-hazard Simulation, Japan Society for Computational Engineering and Science.

Journal Reviewer

- Computers and Geotechnics (JCR Q1)
- Computational Particle Mechanics (JCR Q1)
- Journal of Petroleum Exploration and Production Technology (JCR Q2)
- Journal of Road Engineering (JCR Q1)
- Particuology (JCR Q2)
- Powder Technology (JCR Q2)
- Scientific Reports (JCR Q1)
- Transportation Geotechnics (JCR Q1)
- Other international journals

RESEARCH FUNDING

JST-SPRING Fellowship, Japan Science and Technology Agency (JST)

Oct. 2021 – Mar. 2024

Principal Investigator

SPRING: Support for Pioneering Research Initiated by the Next Generation

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